MNS Department
Assignment - 1
Semester: Spring-2024
Course ID: PHY 111; Course Title: Principles of Physics I
Total Marks: 15

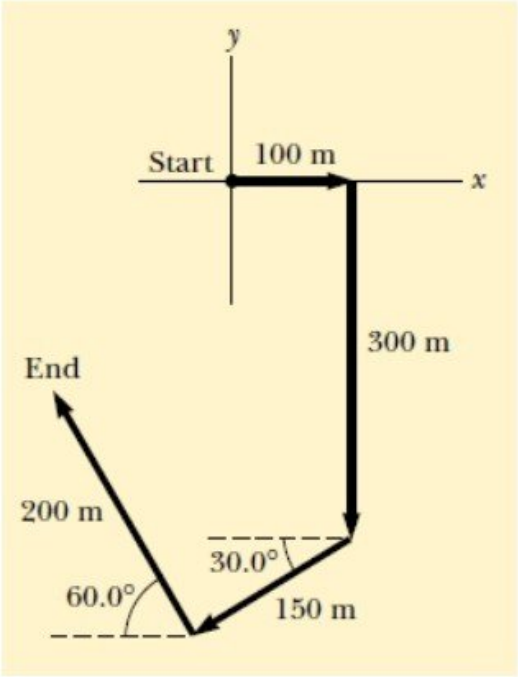

Fig. 1

1. (4 marks) A person going for a walk follows the path shown in Fig. 1. The total trip consists of four straight-line paths. At the end of the walk, what is the person's resultant displacement measured from the starting point?

2. Consider three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$. $\vec{A}$ has co-ordinate $(-3, 2, 4)$, $\vec{B}$ has co-ordinate $(5, -3, -1)$ and $\vec{C}$ has co-ordinate $(-4, 7, 6)$.

   (a) (3 marks) Calculate the angle between $\vec{D}$ and $\vec{B}$ where $\vec{C} = (\vec{D} + \vec{A})$.

   (b) (3 marks) Find $(\vec{B} \times \vec{A})$ and $(\vec{C} \bullet 3\vec{B})$.

3. The acceleration of a car is given by $a(t) = 3.2\ t$. All variables are in SI units.

   (a) (2.5 marks) If the car's velocity at time $t = 2s$ is 8 m/s what is its velocity at time $t = 5s$?

   (b) (2.5 marks) If the car's position at time $t = 2s$ is 10.0 m, what is its average velocity between time $t = 3s$ to $t = 5s$?

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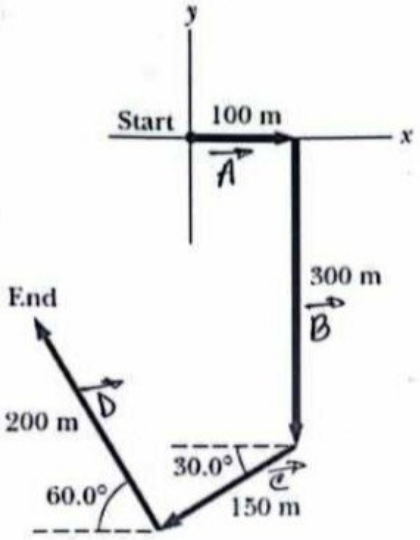

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(1) $\vec{A} = 100\cos 0^\circ\,\hat{i} + 100\sin 0^\circ\,\hat{j} = 100\hat{i}$

$\vec{B} = 300\cos 270^\circ\,\hat{i} + 300\sin 270^\circ\,\hat{j} = -300\hat{j}$

$\vec{C} = 150\cos 210^\circ\,\hat{i} + 150\sin 210^\circ\,\hat{j} = -129.9\hat{i} - 75\hat{j}$

$\vec{D} = 200\cos 120\,\hat{i} + 200\sin 120^\circ\,\hat{j} = -100\hat{i} + 173.21\hat{j}$

Resultant displacement, $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$

$= 100\hat{i} - 300\hat{j} - 129.9\hat{i} - 75\hat{j} - 100\hat{i} + 173.21\hat{j}$

$= \boxed{-129.9\hat{i} - 201.79\hat{j}}$

② $\vec{A} = -3\hat{i}+2\hat{j}+4\hat{k}$, $\vec{B} = 5\hat{i}-3\hat{j}-\hat{k}$, $\vec{C} = -4\hat{i}+7\hat{j}+6\hat{k}$

ⓐ $\vec{C} = \vec{D} + \vec{A}$

$\Rightarrow \vec{D} = \vec{C} - \vec{A}$

$= -4\hat{i}+7\hat{j}+6\hat{k}+3\hat{i}-2\hat{j}-4\hat{k}$

$\therefore \vec{D} = -\hat{i}+5\hat{j}+2\hat{k}$

Again, $\vec{D}\cdot\vec{B} = DB\cos\theta$

$\Rightarrow \theta = \cos^{-1}\left(\frac{\vec{D}\cdot\vec{B}}{DB}\right)$

$= \cos^{-1}\left(\frac{-22}{\sqrt{30}\times\sqrt{35}}\right)$

$\boxed{\therefore \theta = 132.76}$

$\vec{D}\cdot\vec{B} = (-\hat{i}+5\hat{j}+2\hat{k})\cdot(5\hat{i}-3\hat{j}-\hat{k})$

$= -5-15-2$

$= -22$

$D = \sqrt{(-1)^2+5^2+2^2} = \sqrt{30}$

$B = \sqrt{5^2+(-3)^2+(-1)^2} = \sqrt{35}$

ⓑ $\vec{B}\times\vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -3 & -1 \\ -3 & 2 & 4 \end{vmatrix}$

$= \hat{i}(-12+2) - \hat{j}(20-3) + \hat{k}(10-9)$

$= \boxed{-10\hat{i} - 17\hat{j} + \hat{k}}$

and, $\vec{C}\cdot 3\vec{B} = (-4\hat{i}+7\hat{j}+6\hat{k})\cdot(15\hat{i}-9\hat{j}-3\hat{k})$

$= -60-63-18$

$= \boxed{-141}$

③ $a = 3.2t$

$V = \int a\,dt = \int 3.2t\,dt = \frac{3.2t^2}{2} + c_1$

$\therefore V(t) = 1.6t^2 + 1.6$

$x = \int v\,dt = \int(1.6t^2+1.6)\,dt = \frac{1.6t^3}{3} + 1.6t + c_2$

$\therefore x(t) = 0.53t^3 + 1.6t + 2.53$

$V_2 = \frac{3.2\times 2^2}{2} + c_1$

$\Rightarrow 8 = 6.4 + c_1$

$\therefore c_1 = 1.6$

$x_2 = \frac{1.6\times 2^3}{3} + 1.6\times 2 + c_2$

$\Rightarrow 10 = 4.27 + 3.2 + c_2$

$\therefore c_2 = 2.53$

ⓐ $V_5 = 1.6\times 5^2 + 1.6 = \boxed{41.6\ ms^{-1}}$

ⓑ $V_{avg} = \frac{x_5 - x_3}{5-3} = \frac{76.78-21.64}{2}$

$= \boxed{27.57\ ms^{-1}}$

$x_3 = 0.53\times 3^3 + (1.6\times 3) + 2.53$

$= 21.64$

$x_5 = 0.53\times 5^3 + (1.6\times 5) + 2.53$

$= 76.78$